

Isolation Levels



Is ACID too demanding?

Can increase concurrency & reduce locking overhead if weaken guarantees

- Isolation levels
- BASE in some NoSQL

Isolation levels

Isolation level	Dirty reads	Unrepeatable reads	Phantom reads	Default in
Read uncommitted	Allowed	Allowed	Allowed	
Read committed	Not allowed	Some allowed	Some allowed	Most, including PostgreSQL
Repeatable read	Not allowed	Not allowed	Some allowed	MySQL
Serializable	Not allowed	Not allowed	Not allowed	ANSI standard

Not all SQL versions support all four levels.

Syntax

```
BEGIN TRANSACTION;
    SET TRANSACTION ISOLATION LEVEL SERIALIZABLE;
    …
COMMIT TRANSACTION;
```